

SOLUTIONS

1. (1 point) Consider the curve given by $x^2 + xy + y^2 = 7$. At the point $(1, 2)$, is the curve increasing or decreasing?

- A. Increasing
B. Decreasing
C. Neither
D. Cannot be determined

Solution: Differentiate implicitly:

$$2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx}(x + 2y) = -(2x + y) \implies \frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

At $(1, 2)$:

$$\left. \frac{dy}{dx} \right|_{(1,2)} = -\frac{2(1) + 2}{1 + 2(2)} = -\frac{4}{5} < 0.$$

Since $\frac{dy}{dx} < 0$, the curve is **decreasing** at $(1, 2)$.

(b) is correct.

2. (1 point) The cost of producing a cylindrical can of radius r cm is

$$C(r) = \frac{1}{4}r^2 + \frac{27}{2r}, \quad r > 0.$$

At which value of r is the cost minimized?

- A. $\frac{3}{2}$
B. 2
C. 3
D. 9

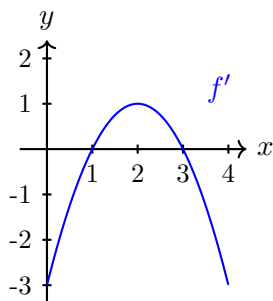
Solution: Differentiate and set to zero:

$$C'(r) = \frac{1}{2}r - \frac{27}{2r^2} = 0 \implies r^3 = 27 \implies r = 3.$$

Since $C''(r) = \frac{1}{2} + \frac{27}{r^3} > 0$ for all $r > 0$, the function is concave up everywhere, so the critical point $r = 3$ is the global minimum.

(c) is correct.

3. (1 point) The graph of f' , the derivative of a twice-differentiable function f , is shown below. On which interval is f both increasing and concave down?



- A. $(1, 2)$
- B. $(2, 3)$
- C. $(3, 4)$
- D. $(0, 1)$

Solution: f is increasing where $f' > 0$ and concave down where f' is decreasing (i.e., $f'' < 0$). From the graph, f' is positive on $(1, 3)$ and decreasing on $(2, 4)$, so (b) is correct.

4. (1 point) A rectangle has its base on the x -axis and its upper two vertices on the parabola $y = 12 - x^2$. Which of the following expressions gives the area of the rectangle as a function of x , where (x, y) is one of the upper vertices?

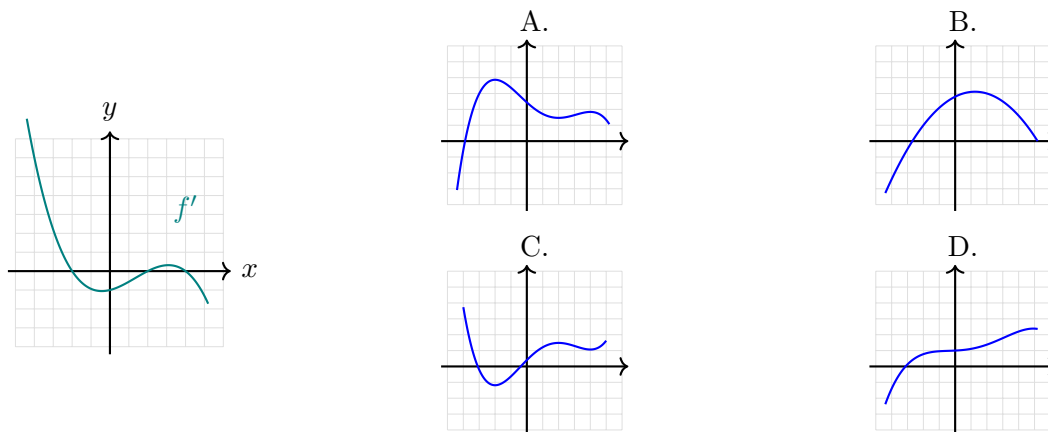
- A. $A(x) = x(12 - x^2)$
- B. $A(x) = 2x(12 - x^2)$
- C. $A(x) = 2x(12 - x)^2$
- D. $A(x) = (2x)^2(12 - x^2)$

Solution: By symmetry, the rectangle has corners $(-x, 0)$, $(x, 0)$, $(x, 12 - x^2)$, $(-x, 12 - x^2)$. The base has length $2x$ and the height is $12 - x^2$.

So $A(x) = 2x(12 - x^2) = 24x - 2x^3$.

(b) is correct.

5. (1 point) The graph of f' , the derivative of a differentiable function f , is shown below. Which of the following is a possible graph of f ?



Solution: We want f to have 0 slope at 2 ticks left of the y -axis, then 2 and 4 ticks right of the y -axis, so B is out (B is if you misread the question and end up finding the graph of the derivative of what's given). $f' > 0$ up to the first root, so f should be increasing up to the 0 derivative, which rules out C. Finally, $f' < 0$ in some areas means f must be decreasing there, ruling out D. A works.